```
      name:  main
       log:  /Users/mhdquan/Documents/thesis-data-analytics/results/analysis_l
> og.smcl
  log type:  smcl
 opened on:  14 May 2026, 00:37:19

1 .
2 . * --- Load panel data ---
3 . use "merged_panel.dta", clear

4 . describe, short

Contains data from merged_panel.dta
 Observations:           219
    Variables:           131                  13 May 2026 23:43
Sorted by:

5 . summarize
```

Variable	Obs	Mean	Std. dev.	Min	Max
id	219	110	63.36403	1	219
source	219	0	0	0	0
N1	0				
gender	219	1.438356	.7416749	0	2
age	219	27.43836	2.993815	23	35
A5_tiktok	219	.5616438	.4973222	0	1
A5_shopee	219	.5296804	.5002618	0	1
A5_lazada	219	.3242009	.4691478	0	1
A5_facebook	219	.2922374	.4558328	0	1
A5_other	219	0	0	0	0
C1	219	1.69863	.5073297	1	3
education	219	2.808219	.8238008	1	4
occupation	219	3.054795	1.291011	1	8
income	217	6.013825	17.0716	1	99
relationship	219	1.931507	.9384073	1	4
D1	219	1.972603	.9904026	1	5
D2	186	2.666667	.9845656	1	4
D3	219	2.812785	1.336372	1	5
D4	219	1.990868	1.381366	0	4
D5	219	3.493151	1.231728	1	5

D6	219	2.835616	1.553455	0	5
EXP1	219	12.21461	10.09062	0	48
EXP2	219	6.431963	5.649534	.7	45
EXP3	219	2.767123	1.224976	1	5
EXP4	219	5.232877	6.197485	0	48
EXP5	219	2.908676	1.271013	1	5
EXP6	219	3.045662	1.099507	1	5
IBT1	219	3.995434	1.272496	1	7
IBT2	219	3.547945	1.204439	1	7
IBT3	219	3.456621	1.173753	1	6
IBT4	218	3.458716	1.251963	1	7
IBT5	218	3.545872	1.319896	1	7
IBT6	219	4.237443	1.103898	1	7
IBT7	216	4.037037	1.271702	1	7
IBT8	219	3.826484	1.199018	1	7
IBT9	217	3.741935	1.204829	1	6
SC1	218	3.788991	1.281369	1	7
SC2	216	4.393519	1.318122	1	7
SC3	219	4.56621	1.410625	1	7
SC4	217	4.248848	1.306257	1	7
SC5	219	4.598174	1.307423	1	7
SC6	218	3.880734	1.278859	1	7
SC7	219	4.799087	1.283558	1	7
SC8	219	3.159817	1.35695	1	7
SC9	218	4.518349	1.230238	1	7
SC10	218	4.509174	1.334549	1	7
SC11	217	4.129032	1.277271	1	7
SC12	219	4.570776	1.259192	2	7
SC13	219	4.447489	1.327456	1	7
IMC1	219	4.940639	.6361737	1	7
SE1	219	4.484018	1.174449	1	7
SE2	219	4.611872	1.145412	2	7
SE3	219	4.726027	1.136471	2	7
SE4	219	4.456621	1.142061	2	7
SII1	219	3.995434	1.322001	1	7
SII2	219	3.808219	1.299162	1	7
SII3	219	3.899543	1.214949	1	7
SII4	219	4.100457	1.211167	1	7
IMC2	219	1.251142	.7517173	1	7

IPB1	219	3.246575	3.546988	0	18
IPB2	219	2.3379	1.500757	0	5
IPB3	219	2.52968	1.670792	0	5
IPB4	219	3.616438	1.312922	1	7
IPB5	219	3.461187	1.33482	1	7
IPB6	219	3.940639	1.292583	1	7
CONSIST1	219	3.703196	1.18032	1	7
MA1	219	3.817352	1.359125	1	7
MA2	219	4.196347	1.378999	1	7
MA3	219	3.789954	1.16197	1	7
TA1	219	3.570776	1.152685	1	7
TA2	219	4.420091	1.276735	1	7
TA3	219	3.666667	1.138699	1	7
MOOD1	219	3.333333	.8743718	1	5
MOOD2	219	3.16895	.8204632	1	5
FOMO1	219	2.990868	1.013627	1	5
FOMO2	219	2.96347	1.022023	1	5
FOMO3	219	2.716895	1.054583	1	5
A2_T2	219	1.913242	.9705564	1	4
A3_T2	219	.8219178	.3834584	0	1
EXP1_T2	218	12.11468	13.13842	0	105
EXP2_T2	219	5.068037	4.273793	.4	29.7
EXP3_T2	219	2.872146	1.292924	1	5
EXP4_T2	219	4.479452	4.826655	0	25
EXP5_T2	219	2.981735	1.215207	1	5
EXP6_T2	219	2.958904	1.208969	1	5
EXP7_T2	219	2.721461	1.602763	1	6
EXP8_T2	219	2.365297	1.130836	1	4
PS1_T2	214	4.471963	1.235872	1	7
PS2_T2	217	4.59447	1.194729	1	7
PS3_T2	218	4.288991	1.310703	1	7
PS4_T2	216	4.069444	1.346759	1	7
PS5_T2	215	4.130233	1.264825	1	7
PSI1_T2	217	3.746544	1.317721	1	7
PSI2_T2	216	4.083333	1.305623	1	7
PSI3_T2	214	3.957944	1.381673	1	7
PSI4_T2	217	4.078341	1.227894	1	7
PSI5_T2	215	3.809302	1.309886	1	7

EA1_T2	215	4.381395	1.133033	2	7
EA2_T2	217	4.059908	1.305652	1	7
EA3_T2	216	3.851852	1.355882	1	7
EA4_T2	213	3.732394	1.284425	1	7
IMC1_T2	219	3.018265	.4589924	1	7
SP1_T2	216	3.925926	1.32035	1	7
SP2_T2	219	4.123288	1.218495	1	7
SP3_T2	219	3.835616	1.330796	1	7
SP4_T2	219	4.086758	1.262282	1	7
MV1_T2	219	4.424658	1.214414	2	7
MV2_T2	219	4.415525	1.416138	1	7
UBI1_T2	219	3.616438	1.168746	1	7
UBI2_T2	219	3.575342	1.225694	1	7
UBI3_T2	219	3.278539	1.06661	1	6
UBI4_T2	219	3.447489	1.145686	1	7
IPB1_T2	219	4.689498	4.653541	0	31
IPB2_T2	219	3.52968	1.560047	1	6
IPB3_T2	219	3.575342	1.64722	1	6
IPB4_T2	219	3.835616	1.459047	1	7
IPB5_T2	219	3.767123	1.276325	1	7
IPB6_T2	219	4.173516	1.373749	1	7
IMC2_T2	219	3.812785	1.426043	1	7
MA1_T2	219	3.821918	1.299356	1	7
MA2_T2	219	4.378995	1.294915	1	7
MA3_T2	219	3.680365	1.191764	1	7
TA1_T2	219	3.625571	1.167796	1	7
TA2_T2	219	4.219178	1.26976	1	7
TA3_T2	219	3.739726	1.219423	1	7
PEER1	219	3.392694	1.327598	1	7
PEER2	219	3.187215	1.210285	1	6
PEER3	219	3.406393	1.134996	1	6
MOOD1_T2	219	3.456621	.9441707	1	5
MOOD2_T2	219	3.3379	.8155469	1	5
L1	219	.3881279	.4884404	0	1

```
. 
. /*-----------------------------------------------------------------------
>    SECTION 1: DATA PREPARATION & COMPOSITE CONSTRUCTION
> -----------------------------------------------------------------------*/
. 
. * 1.1 Reverse-code items
. * SC reverse items: SC2,3,4,5,7,9,10,12,13
. foreach v in SC2 SC3 SC4 SC5 SC7 SC9 SC10 SC12 SC13 {
  2.     gen `v'_r = 8 - `v'
  3. }
(3 missing values generated)
(2 missing values generated)
(1 missing value generated)
(1 missing value generated)

. * IBT reverse item: IBT8
. gen IBT8_r = 8 - IBT8

. * MA reverse: MA2, MA2_T2
. gen MA2_r = 8 - MA2

. gen MA2_T2_r = 8 - MA2_T2

. * TA reverse: TA2, TA2_T2
. gen TA2_r = 8 - TA2

. gen TA2_T2_r = 8 - TA2_T2

. 
. * 1.2 Compute scale composites (T1)
. * IBT mean (using recoded IBT8)
. egen IBT_mean = rowmean(IBT1 IBT2 IBT3 IBT4 IBT5 IBT6 IBT7 IBT8_r IBT9)

. 
```

```
. * SC mean (using recoded reverse items)
. egen SC_mean = rowmean(SC1 SC2_r SC3_r SC4_r SC5_r SC6 SC7_r SC8 SC9_r SC10_
> r SC11 SC12_r SC13_r)

.
. * SE mean
. egen SE_mean = rowmean(SE1 SE2 SE3 SE4)

.
. * SII mean
. egen SII_mean = rowmean(SII1 SII2 SII3 SII4)

.
. * FoMO mean
. egen FOMO_mean = rowmean(FOMO1 FOMO2 FOMO3)

.
. * Money/Time availability (T1)
. egen MA_mean = rowmean(MA1 MA2_r MA3)

. egen TA_mean = rowmean(TA1 TA2_r TA3)

.
. * 1.3 Compute T2 exposure composite index
. * Step 1: z-scores of components
. egen EXP5_T2_z = std(EXP5_T2)

. gen ln_EXP2_T2 = ln(EXP2_T2 + 1)

. egen ln_EXP2_T2_z = std(ln_EXP2_T2)

. egen EXP3_T2_z = std(EXP3_T2)

.
```

```
* Step 2: Weighted composite (0.40, 0.30, 0.30)
gen Exposure_T2 = 0.40 * EXP5_T2_z + 0.30 * ln_EXP2_T2_z + 0.30 * EXP3_T2_z
label var Exposure_T2 "T2 Exposure Composite Index"

* 1.4 Compute T2 mediator composites
egen PS_T2 = rowmean(PS1_T2 PS2_T2 PS3_T2 PS4_T2 PS5_T2)
egen PSI_T2 = rowmean(PSI1_T2 PSI2_T2 PSI3_T2 PSI4_T2 PSI5_T2)
egen EA_T2 = rowmean(EA1_T2 EA2_T2 EA3_T2 EA4_T2)
egen SP_T2 = rowmean(SP1_T2 SP2_T2 SP3_T2 SP4_T2)

* 1.5 Compute T2 robustness DV (Likert composite)
egen IPB_Likert_T2 = rowmean(IPB4_T2 IPB5_T2 IPB6_T2)

* 1.6 T2 time-varying controls
egen MA_T2_mean = rowmean(MA1_T2 MA2_T2_r MA3_T2)
egen TA_T2_mean = rowmean(TA1_T2 TA2_T2_r TA3_T2)

* 1.7 Mean-center moderators for interaction terms
summarize IBT_mean, meanonly
gen IBT_c = IBT_mean - r(mean)
summarize SC_mean, meanonly
gen SC_c = SC_mean - r(mean)
```

```
70 .
71 . * 1.8 Binary exposure for PSM
72 . summarize Exposure_T2, detail

                 T2 Exposure Composite Index
-------------------------------------------------------------
      Percentiles      Smallest
 1%    -1.641271      -1.740312
 5%     -1.50975      -1.641271
10%    -1.167895      -1.641271       Obs                 219
25%     -.697338      -1.584534       Sum of wgt.         219

50%    -.0001338                      Mean          -1.33e-09
                        Largest       Std. dev.      .8953857
75%     .5665714       1.789942
90%     1.309881       1.845443       Variance       .8017156
95%     1.604637       1.876059       Skewness       .1793324
99%     1.845443       1.930052       Kurtosis       2.285757

73 . gen Exposure_binary = (Exposure_T2 >= r(p50))

74 . label var Exposure_binary "High exposure (above median)"

75 .
76 . /*----------------------------------------------------------------------
   >   SECTION 2: DESCRIPTIVE STATISTICS
   > ----------------------------------------------------------------------*/
77 .
78 . * 2.1 Table 1 descriptives
79 . dtable age i.gender i.education i.occupation income i.relationship ///
   >        IBT_mean SC_mean SE_mean SII_mean FOMO_mean ///
   >        IPB1 Exposure_T2 ///
   >        PS_T2 PSI_T2 EA_T2 SP_T2 IPB1_T2, ///
   >        title("Table 1: Descriptive Statistics") ///
   >        export("results/table1_descriptives.docx", replace)
```

Table 1: Descriptive Statistics

	Summary
N	**219**
age	**27.438 (2.994)**
gender	
0	**33 (15.1%)**
1	**57 (26.0%)**
2	**129 (58.9%)**
education	
1	**23 (10.5%)**
2	**30 (13.7%)**
3	**132 (60.3%)**
4	**34 (15.5%)**
occupation	
1	**22 (10.0%)**
2	**23 (10.5%)**
3	**138 (63.0%)**
4	**18 (8.2%)**
5	**7 (3.2%)**
6	**2 (0.9%)**
7	**4 (1.8%)**
8	**5 (2.3%)**
income	**6.014 (17.072)**
relationship	
1	**89 (40.6%)**
2	**71 (32.4%)**
3	**44 (20.1%)**
4	**15 (6.8%)**
IBT_mean	**3.799 (0.854)**
SC_mean	**3.561 (0.852)**
SE_mean	**4.570 (0.917)**
SII_mean	**3.951 (0.989)**
FOMO_mean	**2.890 (0.820)**
IPB1	**3.247 (3.547)**
T2 Exposure Composite Index	**-0.000 (0.895)**
PS_T2	**4.313 (1.012)**
PSI_T2	**3.931 (1.052)**
EA_T2	**4.009 (1.007)**
SP_T2	**3.993 (1.051)**
IPB1_T2	**4.689 (4.654)**

(collection **DTable** exported to file results/table1_descriptives.docx)

```
80 .
81 . * 2.2 Correlation matrix
82 . correlate Exposure_T2 PS_T2 PSI_T2 EA_T2 SP_T2 IPB1_T2 ///
   >          IBT_mean SC_mean IPB1
   (obs=219)

                | Exposu~2    PS_T2   PSI_T2    EA_T2    SP_T2  IPB1_T2 IBT_mean
   >  SC_mean
   -------------+---------------------------------------------------------------
   > ---------
    Exposure_T2 |   1.0000
          PS_T2 |   0.2885   1.0000
         PSI_T2 |   0.2935   0.1343   1.0000
          EA_T2 |   0.4108   0.1674   0.1556   1.0000
          SP_T2 |   0.2976   0.1841   0.0568   0.1827   1.0000
        IPB1_T2 |   0.2194   0.1752   0.1529   0.0692   0.1962   1.0000
       IBT_mean |   0.3597   0.1209   0.0562   0.1982   0.1416   0.0235   1.0000
        SC_mean |  -0.1901   0.0222   0.0524  -0.1289  -0.1585  -0.0905  -0.3904
   >   1.0000
           IPB1 |   0.1187   0.0820   0.0716  -0.1051  -0.0525   0.1861   0.2499
   >  -0.0458

                |     IPB1
   -------------+---------
           IPB1 |   1.0000


83 .
84 . * Export correlation matrix
85 . quietly correlate Exposure_T2 PS_T2 PSI_T2 EA_T2 SP_T2 IPB1_T2 ///
   >                    IBT_mean SC_mean IPB1

86 . matrix C = r(C)
```

```
87 . putexcel set "results/table2_correlations.xlsx", replace
   note: file will be replaced when the first putexcel command is issued.

88 . putexcel A1 = matrix(C), names nformat(number_d2)
   file results/table2_correlations.xlsx saved

89 .
90 . * 2.3 DV distribution check
91 . summarize IPB1_T2, detail

                           IPB1_T2
   -------------------------------------------------------------
         Percentiles      Smallest
    1%            0              0
    5%            0              0
   10%            0              0       Obs                 219
   25%            2              0       Sum of wgt.         219

   50%            3                      Mean           4.689498
                          Largest        Std. dev.      4.653541
   75%            7             19
   90%           11             19       Variance       21.65544
   95%           15             22       Skewness       1.970019
   99%           19             31       Kurtosis        8.48683

92 . histogram IPB1_T2, discrete frequency ///
   >     title("Distribution of IPB1_T2 (Count DV)")
   (start=0, width=1)

93 . graph export "results/fig1_ipb_distribution.png", replace width(1200)
   file
       /Users/mhdquan/Documents/thesis-data-analytics/results/fig1_ipb_distribu
       > tion.png saved as PNG format
```

```
 94 .
 95 . /*-------------------------------------------------------------------------
    >   SECTION 3: MEASUREMENT VALIDATION
    > -----------------------------------------------------------------------*/
 96 .
 97 . * 3.1 Cronbach's Alpha
 98 . display _n "--- Cronbach's Alpha ---"

    --- Cronbach's Alpha ---

 99 .
100 . * IBT (with reverse-coded IBT8)
101 . alpha IBT1 IBT2 IBT3 IBT4 IBT5 IBT6 IBT7 IBT8_r IBT9, item

    Test scale = mean(unstandardized items)

                                                                  Average
                                         Item-test    Item-rest    interitem
    Item          |  Obs  Sign   correlation  correlation   covariance      alph
    > a
    --------------+-------------------------------------------------------------
    > -
    IBT1          |  219    +        0.7417       0.6508      .6172433     0.850
    > 9
    IBT2          |  219    +        0.7047       0.6111        .63751     0.854
    > 8
    IBT3          |  219    +        0.7065       0.6159      .6402384     0.854
    > 4
    IBT4          |  218    +        0.7162       0.6212      .6283658     0.853
    > 8
    IBT5          |  218    +        0.7678       0.6810       .602537     0.847
    > 9
    IBT6          |  219    +        0.7091       0.6252      .6473783     0.854
    > 0
    IBT7          |  216    +        0.7164       0.6200      .6257515     0.853
    > 8
    IBT8_r        |  219    +        0.5218       0.3940      .6988278     0.873
    > 9
    IBT9          |  217    +        0.7043       0.6095       .637429     0.854
    > 8
    --------------+-------------------------------------------------------------
    > -
    Test scale    |                                           .6372435     0.869
    > 5
    --------------+-------------------------------------------------------------
    > -
```

```
102 .
103 . * SC (with reverse-coded items)
104 . alpha SC1 SC2_r SC3_r SC4_r SC5_r SC6 SC7_r SC8 SC9_r SC10_r SC11 SC12_r SC1
    > 3_r, item

    Test scale = mean(unstandardized items)

                                                                   Average
                                    Item-test      Item-rest      interitem
    Item           |  Obs  Sign   correlation   correlation     covariance       alph
    > a
    ---------------+--------------------------------------------------------------
    > -
    SC1            |  218    +        0.7373        0.6767        .631056       0.875
    > 1
    SC2_r          |  216    +        0.6633        0.5879       .6442746       0.879
    > 5
    SC3_r          |  219    +        0.6542        0.5710       .6399962       0.880
    > 5
    SC4_r          |  217    +        0.5882        0.5026       .6609813       0.883
    > 5
    SC5_r          |  219    +        0.6817        0.6097       .6410645       0.878
    > 4
    SC6            |  218    +        0.6372        0.5602       .6528049       0.880
    > 9
    SC7_r          |  219    +        0.6167        0.5359       .6570107       0.882
    > 0
    SC8            |  219    +        0.6320        0.5489       .6489117       0.881
    > 6
    SC9_r          |  218    +        0.6724        0.6042       .6485401       0.878
    > 7
    SC10_r         |  218    +        0.6405        0.5606       .6478865       0.880
    > 8
    SC11           |  217    +        0.6329        0.5558       .6535218       0.881
    > 0
    SC12_r         |  219    +        0.6883        0.6206       .6430183       0.877
    > 9
    SC13_r         |  219    +        0.6536        0.5756        .645846       0.880
    > 1
    ---------------+--------------------------------------------------------------
    > -
    Test scale     |                                              .6473023       0.888
    > 2
    ---------------+--------------------------------------------------------------
    > -
```

```
105 .
106 . * SE
107 . alpha SE1 SE2 SE3 SE4, item

    Test scale = mean(unstandardized items)

                                                                    Average
                                    Item-test      Item-rest       interitem
    Item         |  Obs  Sign     correlation   correlation      covariance       alph
    > a
    -------------+----------------------------------------------------------------------
    > -
    SE1          |  219    +         0.8025         0.6285          .668077       0.759
    > 5
    SE2          |  219    +         0.8416         0.6999         .6193219       0.724
    > 7
    SE3          |  219    +         0.7602         0.5697         .7347898       0.786
    > 9
    SE4          |  219    +         0.7854         0.6080         .6986162       0.769
    > 2
    -------------+----------------------------------------------------------------------
    > -
    Test scale   |                                                 .6802012       0.809
    > 2
    -------------+----------------------------------------------------------------------
    > -

108 .
109 . * SII
110 . alpha SII1 SII2 SII3 SII4, item

    Test scale = mean(unstandardized items)

                                                                    Average
                                    Item-test      Item-rest       interitem
    Item         |  Obs  Sign     correlation   correlation      covariance       alph
    > a
    -------------+----------------------------------------------------------------------
    > -
    SII1         |  219    +         0.7985         0.6109         .7373242       0.732
    > 9
    SII2         |  219    +         0.8018         0.6211         .7357952       0.727
    > 3
    SII3         |  219    +         0.7810         0.6046         .7876573       0.736
    > 3
```

```
SII4          │  219    +      0.7525       0.5611        .8340804      0.757
> 0
──────────────┼──────────────────────────────────────────────────────────────
> ─
Test scale    │                                          .7737143      0.790
> 4
──────────────┴──────────────────────────────────────────────────────────────
> ─

111 .
112 . * PS
113 . alpha PS1_T2 PS2_T2 PS3_T2 PS4_T2 PS5_T2, item

Test scale = mean(unstandardized items)

                                                          Average
                              Item-test     Item-rest     interitem
Item          │  Obs  Sign   correlation   correlation    covariance      alph
> a
──────────────┼──────────────────────────────────────────────────────────────
> ─
PS1_T2        │  214    +      0.7972       0.6716        .8820468      0.823
> 2
PS2_T2        │  217    +      0.7639       0.6322        .9221218      0.832
> 3
PS3_T2        │  218    +      0.8006       0.6685        .8466945      0.819
> 6
PS4_T2        │  216    +      0.8042       0.6705        .8377271      0.821
> 0
PS5_T2        │  215    +      0.8077       0.6862        .8570289      0.817
> 6
──────────────┼──────────────────────────────────────────────────────────────
> ─
Test scale    │                                          .8691265      0.853
> 1
──────────────┴──────────────────────────────────────────────────────────────
> ─
```

```
114 .
115 . * PSI
116 . alpha PSI1_T2 PSI2_T2 PSI3_T2 PSI4_T2 PSI5_T2, item

    Test scale = mean(unstandardized items)

                                                                    Average
                                       Item-test     Item-rest      interitem
    Item          │   Obs  Sign   correlation   correlation      covariance      alph
    > a
    ──────────────┼──────────────────────────────────────────────────────────────────
    > -
    PSI1_T2       │   217    +        0.7982        0.6694        .9427029      0.831
    > 2
    PSI2_T2       │   216    +        0.8220        0.7075        .9299581      0.825
    > 4
    PSI3_T2       │   214    +        0.7862        0.6426         .948377      0.840
    > 9
    PSI4_T2       │   217    +        0.7888        0.6647        .9876851      0.835
    > 4
    PSI5_T2       │   215    +        0.8205        0.7052        .9260077      0.824
    > 6
    ──────────────┼──────────────────────────────────────────────────────────────────
    > -
    Test scale    │                                              .9469205      0.860
    > 5
    ──────────────┴──────────────────────────────────────────────────────────────────
    > -

117 .
118 . * EA
119 . alpha EA1_T2 EA2_T2 EA3_T2 EA4_T2, item

    Test scale = mean(unstandardized items)
```

```
                                                         Average
                                Item-test    Item-rest   interitem
Item           |  Obs   Sign   correlation  correlation   covariance      alph
> a
---------------+------------------------------------------------------------------
> -
EA1_T2         |  215     +       0.7940       0.6419      .8461613      0.741
> 5
EA2_T2         |  217     +       0.8030       0.6197      .8003136      0.752
> 3
EA3_T2         |  216     +       0.8250       0.6421      .7423286      0.734
> 8
EA4_T2         |  213     +       0.7530       0.5441      .8803194      0.783
> 6
---------------+------------------------------------------------------------------
> -
Test scale     |                                          .8174659      0.803
> 0
---------------+------------------------------------------------------------------
> -

120 .
121 . * SP
122 . alpha SP1_T2 SP2_T2 SP3_T2 SP4_T2, item

Test scale = mean(unstandardized items)

                                                         Average
                                Item-test    Item-rest   interitem
Item           |  Obs   Sign   correlation  correlation   covariance      alph
> a
---------------+------------------------------------------------------------------
> -
SP1_T2         |  216     +       0.8357       0.6896      .8856562      0.784
> 3
SP2_T2         |  219     +       0.8318       0.6978      .9273749      0.782
> 1
SP3_T2         |  219     +       0.8287       0.6735      .9014505      0.793
> 2
SP4_T2         |  219     +       0.7837       0.6140      .9996487      0.818
> 2
---------------+------------------------------------------------------------------
> -
Test scale     |                                           .928434      0.837
> 7
---------------+------------------------------------------------------------------
```

```
> -

.
. * 3.2 Factor analysis (CFA proxy via EFA)
. * T2 Mediators - check 4-factor structure
. factor PS1_T2 PS2_T2 PS3_T2 PS4_T2 PS5_T2 ///
>        PSI1_T2 PSI2_T2 PSI3_T2 PSI4_T2 PSI5_T2 ///
>        EA1_T2 EA2_T2 EA3_T2 EA4_T2 ///
>        SP1_T2 SP2_T2 SP3_T2 SP4_T2, ///
>        factors(4) pcf
(obs=175)

Factor analysis/correlation                        Number of obs    =        175
    Method: principal-component factors            Retained factors =          4
    Rotation: (unrotated)                          Number of params =         66
```

Factor	Eigenvalue	Difference	Proportion	Cumulative
Factor1	**4.17113**	**1.21304**	**0.2317**	**0.2317**
Factor2	**2.95809**	**0.50482**	**0.1643**	**0.3961**
Factor3	**2.45327**	**0.37882**	**0.1363**	**0.5324**
Factor4	**2.07445**	**1.39424**	**0.1152**	**0.6476**
Factor5	**0.68022**	**0.03938**	**0.0378**	**0.6854**
Factor6	**0.64084**	**0.05932**	**0.0356**	**0.7210**
Factor7	**0.58151**	**0.00924**	**0.0323**	**0.7533**
Factor8	**0.57227**	**0.05674**	**0.0318**	**0.7851**
Factor9	**0.51553**	**0.04819**	**0.0286**	**0.8137**
Factor10	**0.46734**	**0.01448**	**0.0260**	**0.8397**
Factor11	**0.45286**	**0.00899**	**0.0252**	**0.8649**
Factor12	**0.44387**	**0.03510**	**0.0247**	**0.8895**
Factor13	**0.40877**	**0.03228**	**0.0227**	**0.9122**
Factor14	**0.37649**	**0.03726**	**0.0209**	**0.9331**
Factor15	**0.33923**	**0.01880**	**0.0188**	**0.9520**
Factor16	**0.32044**	**0.02788**	**0.0178**	**0.9698**
Factor17	**0.29255**	**0.04143**	**0.0163**	**0.9860**
Factor18	**0.25113**	**.**	**0.0140**	**1.0000**

LR test: independent vs. saturated: chi2(**153**) = **1264.59** Prob>chi2 = **0.0000**

Factor loadings (pattern matrix) and unique variances

Variable	Factor1	Factor2	Factor3	Factor4	Uniqueness
PS1_T2	**0.6633**	**0.2101**	**-0.3242**	**-0.2665**	**0.3398**
PS2_T2	**0.5590**	**0.2463**	**-0.3769**	**-0.2239**	**0.4346**
PS3_T2	**0.5661**	**0.2770**	**-0.4531**	**-0.2525**	**0.3338**
PS4_T2	**0.5492**	**0.1595**	**-0.5252**	**-0.1697**	**0.3683**
PS5_T2	**0.6017**	**0.1594**	**-0.4587**	**-0.2095**	**0.3583**
PSI1_T2	**0.5011**	**-0.5970**	**0.2399**	**0.0052**	**0.3350**
PSI2_T2	**0.5032**	**-0.6092**	**0.2219**	**-0.1130**	**0.3136**
PSI3_T2	**0.4843**	**-0.5613**	**0.2003**	**-0.0917**	**0.4018**
PSI4_T2	**0.5304**	**-0.6058**	**0.1257**	**-0.0343**	**0.3347**
PSI5_T2	**0.4703**	**-0.6303**	**0.1568**	**-0.0788**	**0.3508**
EA1_T2	**0.4751**	**0.1584**	**-0.0171**	**0.6399**	**0.3394**
EA2_T2	**0.4135**	**0.2054**	**0.0791**	**0.6633**	**0.3405**
EA3_T2	**0.4765**	**0.1752**	**0.0777**	**0.6540**	**0.3084**
EA4_T2	**0.4380**	**0.2237**	**0.1102**	**0.5022**	**0.4937**
SP1_T2	**0.3056**	**0.5130**	**0.5566**	**-0.2027**	**0.2926**
SP2_T2	**0.3273**	**0.4585**	**0.5997**	**-0.1759**	**0.2921**
SP3_T2	**0.3735**	**0.4745**	**0.4745**	**-0.3006**	**0.3199**
SP4_T2	**0.1940**	**0.2689**	**0.6388**	**-0.3105**	**0.3856**

```
127 . rotate, varimax

    Factor analysis/correlation                        Number of obs    =        175
        Method: principal-component factors            Retained factors =          4
        Rotation: orthogonal varimax (Kaiser off)      Number of params =         66
```

Factor	Variance	Difference	Proportion	Cumulative
Factor1	**3.27509**	**0.11534**	**0.1819**	**0.1819**
Factor2	**3.15975**	**0.46161**	**0.1755**	**0.3575**
Factor3	**2.69814**	**0.17417**	**0.1499**	**0.5074**
Factor4	**2.52397**	.	**0.1402**	**0.6476**

LR test: independent vs. saturated: chi2(**153**) = **1264.59** Prob>chi2 = **0.0000**

Rotated factor loadings (pattern matrix) and unique variances

Variable	Factor1	Factor2	Factor3	Factor4	Uniqueness
PS1_T2	**0.1365**	**0.7800**	**0.1615**	**0.0841**	**0.3398**
PS2_T2	**0.0323**	**0.7408**	**0.0955**	**0.0796**	**0.4346**
PS3_T2	**-0.0058**	**0.8112**	**0.0673**	**0.0594**	**0.3338**
PS4_T2	**0.0490**	**0.7839**	**-0.0784**	**0.0932**	**0.3683**
PS5_T2	**0.1000**	**0.7901**	**-0.0005**	**0.0862**	**0.3583**
PSI1_T2	**0.8077**	**0.0062**	**0.0208**	**0.1102**	**0.3350**
PSI2_T2	**0.8255**	**0.0565**	**0.0422**	**0.0046**	**0.3136**
PSI3_T2	**0.7695**	**0.0636**	**0.0379**	**0.0239**	**0.4018**
PSI4_T2	**0.8034**	**0.1074**	**-0.0448**	**0.0791**	**0.3347**
PSI5_T2	**0.8027**	**0.0580**	**-0.0376**	**0.0108**	**0.3508**
EA1_T2	**0.0668**	**0.1385**	**-0.0051**	**0.7981**	**0.3394**
EA2_T2	**0.0202**	**0.0438**	**0.0625**	**0.8082**	**0.3405**
EA3_T2	**0.0789**	**0.0801**	**0.0683**	**0.8212**	**0.3084**
EA4_T2	**0.0440**	**0.1013**	**0.1566**	**0.6853**	**0.4937**
SP1_T2	**-0.0623**	**0.0634**	**0.8292**	**0.1089**	**0.2926**
SP2_T2	**0.0014**	**0.0261**	**0.8304**	**0.1327**	**0.2921**
SP3_T2	**-0.0075**	**0.1815**	**0.8035**	**0.0396**	**0.3199**
SP4_T2	**0.1014**	**-0.0900**	**0.7677**	**-0.0817**	**0.3856**

Factor rotation matrix

	Factor1	Factor2	Factor3	Factor4
Factor1	**0.5476**	**0.6444**	**0.3027**	**0.4396**
Factor2	**-0.7843**	**0.2775**	**0.5097**	**0.2193**
Factor3	**0.2736**	**-0.6222**	**0.7303**	**0.0683**
Factor4	**-0.1007**	**-0.3474**	**-0.3395**	**0.8683**

```
128 . estat kmo

    Kaiser-Meyer-Olkin measure of sampling adequacy
```

Variable	kmo
PS1_T2	**0.8443**
PS2_T2	**0.8020**
PS3_T2	**0.8394**
PS4_T2	**0.8074**
PS5_T2	**0.8212**
PSI1_T2	**0.8184**
PSI2_T2	**0.7966**
PSI3_T2	**0.8432**
PSI4_T2	**0.8486**
PSI5_T2	**0.8362**
EA1_T2	**0.7991**
EA2_T2	**0.7510**
EA3_T2	**0.7986**
EA4_T2	**0.8718**
SP1_T2	**0.7933**
SP2_T2	**0.7847**
SP3_T2	**0.7796**
SP4_T2	**0.7634**
Overall	**0.8123**

```
129 .
130 . /*-----------------------------------------------------------------------
    >   SECTION 4: H1 - PRIMARY REGRESSION (Total Effect)
    > -----------------------------------------------------------------------*/
131 .
```

```
132 . display _n "=== H1: Total Effect of Exposure on IPB ==="
```

=== H1: Total Effect of Exposure on IPB ===

```
133 .
134 . * 4.1 Primary specification: ANCOVA with robust SE
135 . regress IPB1_T2 Exposure_T2 IPB1 ///
>         IBT_mean SC_mean SE_mean age i.gender ///
>         MA_T2_mean TA_T2_mean MOOD1_T2, ///
>         vce(robust)
```

Linear regression

Number of obs	=	**219**
F(11, 207)	=	**2.87**
Prob > F	=	**0.0016**
R-squared	=	**0.1100**
Root MSE	=	**4.5054**

IPB1_T2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
Exposure_T2	**1.197172**	**.4746115**	**2.52**	**0.012**	**.26148**	**2.132864**
IPB1	**.2582819**	**.0925977**	**2.79**	**0.006**	**.0757263**	**.4408374**
IBT_mean	**-.6813771**	**.6085667**	**-1.12**	**0.264**	**-1.881161**	**.5184063**
SC_mean	**-.4370713**	**.4581464**	**-0.95**	**0.341**	**-1.340303**	**.4661599**
SE_mean	**-.2392481**	**.3078399**	**-0.78**	**0.438**	**-.8461515**	**.3676554**
age	**.0370874**	**.0869615**	**0.43**	**0.670**	**-.1343564**	**.2085312**
gender						
1	**-.6819876**	**.9372065**	**-0.73**	**0.468**	**-2.529681**	**1.165706**
2	**-.805272**	**.8165161**	**-0.99**	**0.325**	**-2.415026**	**.8044816**
MA_T2_mean	**.5599323**	**.2806497**	**2.00**	**0.047**	**.0066342**	**1.11323**
TA_T2_mean	**-.0702036**	**.277145**	**-0.25**	**0.800**	**-.6165922**	**.4761851**
MOOD1_T2	**-.0480141**	**.3193309**	**-0.15**	**0.881**	**-.6775719**	**.5815438**
_cons	**7.074472**	**3.782876**	**1.87**	**0.063**	**-.383432**	**14.53238**

```
136 . estimates store h1_primary

137 .
138 . * Check VIF
139 . estat vif

    Variable |       VIF       1/VIF  
-------------+----------------------
 Exposure_T2 |      1.26    0.795986
        IPB1 |      1.11    0.902336
    IBT_mean |      1.49    0.669202
     SC_mean |      1.24    0.808774
     SE_mean |      1.26    0.792265
         age |      1.06    0.944518
      gender |
          1  |      2.06    0.485351
          2  |      2.12    0.472620
  MA_T2_mean |      1.05    0.954458
  TA_T2_mean |      1.05    0.950412
    MOOD1_T2 |      1.04    0.962644
-------------+----------------------
    Mean VIF |      1.34

140 .
141 . * 4.2 Robustness: Likert composite DV
142 . regress IPB_Likert_T2 Exposure_T2 IPB1 ///
>         IBT_mean SC_mean SE_mean age i.gender ///
>         MA_T2_mean TA_T2_mean MOOD1_T2, ///
>         vce(robust)

Linear regression                               Number of obs     =        219
                                                F(11, 207)        =       7.30
                                                Prob > F          =     0.0000
                                                R-squared         =     0.2446
                                                Root MSE          =     1.0168
```

```
------------------------------------------------------------------------------
             |               Robust
IPB_Likert~2 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 Exposure_T2 |   .4332369    .088071     4.92   0.000     .2596059     .606868
        IPB1 |    .055615   .0215756     2.58   0.011      .013079    .0981511
    IBT_mean |   .2220996    .096238     2.31   0.022     .0323673    .4118318
     SC_mean |   .0036068   .0915565     0.04   0.969    -.1768958    .1841095
     SE_mean |   -.075131    .081559    -0.92   0.358    -.2359238    .0856617
         age |  -.0061757   .0247468    -0.25   0.803    -.0549637    .0426123
             |
      gender |
          1  |  -.1697781   .2364061    -0.72   0.473    -.6358504    .2962943
          2  |  -.1484653   .2166022    -0.69   0.494    -.5754944    .2785639
             |
  MA_T2_mean |   .1300016   .0624713     2.08   0.039       .00684    .2531632
  TA_T2_mean |  -.0804168   .0689584    -1.17   0.245    -.2163676    .0555341
   MOOD1_T2  |    .158794   .0698469     2.27   0.024     .0210916    .2964965
       _cons |   2.800455   .9367606     2.99   0.003     .9536401    4.647269
------------------------------------------------------------------------------

143 . estimates store h1_likert

144 .
145 . * 4.3 Robustness: Poisson regression for count DV
146 . poisson IPB1_T2 Exposure_T2 IPB1 ///
    >         IBT_mean SC_mean SE_mean age i.gender ///
    >         MA_T2_mean TA_T2_mean MOOD1_T2, ///
    >         vce(robust)

    Iteration 0:  Log pseudolikelihood =  -701.1757
    Iteration 1:  Log pseudolikelihood = -701.17517
    Iteration 2:  Log pseudolikelihood = -701.17517

    Poisson regression                                      Number of obs =    219
                                                            Wald chi2(11) =  39.06
                                                            Prob > chi2   = 0.0001
    Log pseudolikelihood = -701.17517                       Pseudo R2     = 0.0703
```

IPB1_T2	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
Exposure_T2	**.2498297**	**.0888418**	**2.81**	**0.005**	**.075703**	**.4239564**
IPB1	**.0472586**	**.0146909**	**3.22**	**0.001**	**.0184649**	**.0760523**
IBT_mean	**-.1348521**	**.1165796**	**-1.16**	**0.247**	**-.3633439**	**.0936397**
SC_mean	**-.0835922**	**.08681**	**-0.96**	**0.336**	**-.2537368**	**.0865523**
SE_mean	**-.0526502**	**.0685495**	**-0.77**	**0.442**	**-.1870048**	**.0817044**
age	**.0084463**	**.0171148**	**0.49**	**0.622**	**-.025098**	**.0419907**
gender						
1	**-.1551928**	**.1885244**	**-0.82**	**0.410**	**-.5246938**	**.2143082**
2	**-.1946265**	**.1599962**	**-1.22**	**0.224**	**-.5082132**	**.1189602**
MA_T2_mean	**.1225913**	**.0561826**	**2.18**	**0.029**	**.0124754**	**.2327072**
TA_T2_mean	**-.0188258**	**.0575679**	**-0.33**	**0.744**	**-.1316568**	**.0940051**
MOOD1_T2	**-.0004001**	**.0645166**	**-0.01**	**0.995**	**-.1268503**	**.1260501**
_cons	**1.932841**	**.7308209**	**2.64**	**0.008**	**.5004584**	**3.365224**

```
147 . estimates store h1_poisson

148 .
149 . * 4.4 Robustness: Negative binomial
150 . nbreg IPB1_T2 Exposure_T2 IPB1 ///
    >       IBT_mean SC_mean SE_mean age i.gender ///
    >       MA_T2_mean TA_T2_mean MOOD1_T2, ///
    >       vce(robust)

    Fitting Poisson model:

    Iteration 0:  Log pseudolikelihood =  -701.1757
    Iteration 1:  Log pseudolikelihood = -701.17517
    Iteration 2:  Log pseudolikelihood = -701.17517

    Fitting constant-only model:

    Iteration 0:  Log pseudolikelihood = -579.27371
    Iteration 1:  Log pseudolikelihood = -574.89437
    Iteration 2:  Log pseudolikelihood = -574.89336
    Iteration 3:  Log pseudolikelihood = -574.89336

    Fitting full model:
```

```
Iteration 0:  Log pseudolikelihood = -563.97343
Iteration 1:  Log pseudolikelihood = -562.38789
Iteration 2:  Log pseudolikelihood = -562.35009
Iteration 3:  Log pseudolikelihood = -562.35007

Negative binomial regression                            Number of obs =    219
                                                        Wald chi2(11) =  33.66
Dispersion: mean                                        Prob > chi2   = 0.0004
Log pseudolikelihood = -562.35007                       Pseudo R2     = 0.0218
```

IPB1_T2	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
Exposure_T2	.2340151	.080241	2.92	0.004	.0767457	.3912845
IPB1	.0526711	.0144074	3.66	0.000	.024433	.0809091
IBT_mean	-.0836248	.1000972	-0.84	0.403	-.2798117	.112562
SC_mean	-.0314595	.0860128	-0.37	0.715	-.2000416	.1371226
SE_mean	-.0852522	.0721378	-1.18	0.237	-.2266398	.0561354
age	.0056015	.0178882	0.31	0.754	-.0294587	.0406616
gender						
1	-.1168296	.1954934	-0.60	0.550	-.4999897	.2663305
2	-.1944564	.1625075	-1.20	0.231	-.5129653	.1240525
MA_T2_mean	.1293183	.0571574	2.26	0.024	.0172917	.2413448
TA_T2_mean	-.0447132	.0542933	-0.82	0.410	-.1511262	.0616997
MOOD1_T2	-.005589	.0665499	-0.08	0.933	-.1360245	.1248464
_cons	1.842629	.818494	2.25	0.024	.2384102	3.446848
/lnalpha	-.5515078	.1475934			-.8407856	-.26223
alpha	.5760805	.0850257			.4313715	.769334

```
151 . estimates store h1_nbreg

152 .
153 . * Compare specifications
154 . estimates table h1_primary h1_likert h1_poisson h1_nbreg, ///
    >     star(0.10 0.05 0.01) stats(N r2 aic bic)
```

Variable	h1_primary	h1_likert	h1_poisson	h1_nbreg
_				
Exposure_T2	**1.1971719****	**.43323693*****		
IPB1	**.25828185*****	**.05561502****		
IBT_mean	**-.6813771**	**.22209958****		
SC_mean	**-.43707132**	**.00360684**		
SE_mean	**-.23924806**	**-.07513104**		
age	**.03708739**	**-.00617566**		
gender				
1	**-.68198764**	**-.16977808**		
2	**-.80527203**	**-.14846526**		
MA_T2_mean	**.55993227****	**.13000157****		
TA_T2_mean	**-.07020357**	**-.08041678**		
MOOD1_T2	**-.04801406**	**.15879404****		
_cons	**7.0744719***	**2.8004547*****		
IPB1_T2				
Exposure_T2			**.24982968*****	**.23401513*****
IPB1			**.04725862*****	**.05267106*****
IBT_mean			**-.13485211**	**-.08362485**
SC_mean			**-.08359225**	**-.03145948**
SE_mean			**-.0526502**	**-.0852522**
age			**.00844635**	**.00560147**
gender				
1			**-.15519278**	**-.11682964**
2			**-.1946265**	**-.19445642**
MA_T2_mean			**.1225913****	**.12931826****
TA_T2_mean			**-.01882583**	**-.04471322**
MOOD1_T2			**-.00040012**	**-.00558905**
_cons			**1.9328412*****	**1.8426289****
/lnalpha				**-.5515078*****

```
Statistics
              N |        219          219          219          219
             r2 | .10996642    .24461991
            aic | 1292.4614    640.46518    1426.3503    1150.7001
            bic | 1333.1302    681.13404    1467.0192    1194.7581
----------------------------------------------------------------------------
                                   Legend: * p<.1; ** p<.05; *** p<.01

155 .
156 . * Export H1 table
157 . esttab h1_primary h1_likert h1_poisson h1_nbreg ///
    >     using "results/table3_H1_total_effect.csv", replace ///
    >     star(* 0.10 ** 0.05 *** 0.01) ///
    >     b(%9.3f) se(%9.3f) ///
    >     scalars("N Observations" "r2 R-squared" "aic AIC" "bic BIC") ///
    >     title("Table 3: H1 – Total Effect of Exposure on IPB") ///
    >     mtitles("OLS" "Likert DV" "Poisson" "NegBin") ///
    >     note("Robust standard errors in parentheses")
    (file results/table3_H1_total_effect.csv not found)
    (output written to results/table3_H1_total_effect.csv)

158 .
159 . /*-----------------------------------------------------------------
    >   SECTION 5: H2a-d – MEDIATION ANALYSIS (Parallel Multiple Mediator)
    > -----------------------------------------------------------------*/
160 .
161 . display _n "=== H2: Parallel Mediation ==="

    === H2: Parallel Mediation ===

162 .
163 . * 5.1 a-paths: Exposure → each mediator
164 . foreach med in PS_T2 PSI_T2 EA_T2 SP_T2 {
      2.     regress `med' Exposure_T2 IPB1 IBT_mean SC_mean age i.gender, vce(rob
    > ust)
      3.     estimates store a_`med'
      4. }

    Linear regression                               Number of obs     =        219
                                                    F(7, 211)         =       3.34
                                                    Prob > F          =     0.0021
                                                    R-squared         =     0.1083
                                                    Root MSE          =     .97141
```

PS_T2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
Exposure_T2	.3201085	.0803136	3.99	0.000	.1617886	.4784283
IPB1	.0110619	.018876	0.59	0.558	-.0261478	.0482717
IBT_mean	.0551257	.0936524	0.59	0.557	-.1294885	.2397399
SC_mean	.1024043	.0888637	1.15	0.250	-.07277	.2775786
age	-.0143837	.0244351	-0.59	0.557	-.0625519	.0337845
gender						
1	-.3320676	.2451511	-1.35	0.177	-.8153267	.1511915
2	-.0790136	.2271808	-0.35	0.728	-.5268483	.3688212
_cons	4.230614	.9432142	4.49	0.000	2.371284	6.089945

Linear regression

Number of obs	=	219
F(7, 211)	=	4.11
Prob > F	=	0.0003
R-squared	=	0.1109
Root MSE	=	1.0082

PSI_T2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
Exposure_T2	.3848478	.0836373	4.60	0.000	.2199759	.5497196
IPB1	.0112895	.0204781	0.55	0.582	-.0290784	.0516574
IBT_mean	-.0260059	.0945607	-0.28	0.784	-.2124108	.1603989
SC_mean	.1234448	.1017725	1.21	0.227	-.0771764	.324066
age	-.0356121	.0216406	-1.65	0.101	-.0782717	.0070474
gender						
1	.0236287	.2301305	0.10	0.918	-.4300207	.4772782
2	.0590392	.2290392	0.26	0.797	-.3924592	.5105375
_cons	4.489812	.8343	5.38	0.000	2.845181	6.134443

Linear regression

Number of obs	=	219
F(7, 211)	=	10.91
Prob > F	=	0.0000
R-squared	=	0.2238
Root MSE	=	.90148

EA_T2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
Exposure_T2	.4490342	.0694755	6.46	0.000	.3120792	.5859892
IPB1	-.0526292	.0152458	-3.45	0.001	-.0826829	-.0225756
IBT_mean	.1160912	.0795804	1.46	0.146	-.0407833	.2729658
SC_mean	-.0444704	.0838046	-0.53	0.596	-.2096719	.1207311
age	-.0447508	.020598	-2.17	0.031	-.0853551	-.0041465
gender						
1	-.036685	.2104789	-0.17	0.862	-.4515958	.3782259
2	.1365581	.1966925	0.69	0.488	-.2511761	.5242922
_cons	5.054284	.7776711	6.50	0.000	3.521283	6.587284

Linear regression

Number of obs	=	219
F(7, 211)	=	4.72
Prob > F	=	0.0001
R-squared	=	0.1091
Root MSE	=	1.0083

SP_T2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
Exposure_T2	.3300545	.0754996	4.37	0.000	.1812242	.4788847
IPB1	-.0282112	.0197787	-1.43	0.155	-.0672003	.0107779
IBT_mean	.0304757	.0848972	0.36	0.720	-.1368796	.197831
SC_mean	-.1191945	.0871929	-1.37	0.173	-.2910754	.0526863
age	.0081946	.0238943	0.34	0.732	-.0389076	.0552969
gender						
1	-.0046909	.2363583	-0.02	0.984	-.4706171	.4612354
2	-.0496382	.2258966	-0.22	0.826	-.4949415	.395665
_cons	4.198652	.8255991	5.09	0.000	2.571173	5.826131

```
165 .
166 . * 5.2 b-paths: All mediators + exposure → IPB (outcome model)
167 . regress IPB1_T2 Exposure_T2 PS_T2 PSI_T2 EA_T2 SP_T2 ///
    >         IPB1 IBT_mean SC_mean age i.gender ///
    >         MA_T2_mean TA_T2_mean MOOD1_T2, ///
    >         vce(robust)

    Linear regression                               Number of obs     =        219
                                                    F(14, 204)        =       2.73
                                                    Prob > F          =     0.0010
                                                    R-squared         =     0.1508
                                                    Root MSE          =     4.4331
```

IPB1_T2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
Exposure_T2	.6101052	.5307887	1.15	0.252	-.4364301	1.65664
PS_T2	.4849144	.2684285	1.81	0.072	-.0443356	1.014164
PSI_T2	.3598467	.3136764	1.15	0.253	-.2586167	.9783101
EA_T2	-.0061858	.3215697	-0.02	0.985	-.6402123	.6278407
SP_T2	.7146951	.2684565	2.66	0.008	.1853899	1.244
IPB1	.2600853	.0924886	2.81	0.005	.0777291	.4424415
IBT_mean	-.7883881	.5734384	-1.37	0.171	-1.919014	.3422379
SC_mean	-.467123	.4528097	-1.03	0.303	-1.35991	.4256643
age	.0537967	.0934328	0.58	0.565	-.1304212	.2380145
gender						
1	-.5680537	.9412141	-0.60	0.547	-2.423809	1.287701
2	-.8081494	.8296026	-0.97	0.331	-2.443844	.8275456
MA_T2_mean	.5455801	.2962749	1.84	0.067	-.0385736	1.129734
TA_T2_mean	-.1054216	.2900841	-0.36	0.717	-.6773691	.4665259
MOOD1_T2	-.1721919	.3184084	-0.54	0.589	-.7999853	.4556015
_cons	.2809699	4.547715	0.06	0.951	-8.685583	9.247522

```
168 . estimates store outcome_model

169 .
170 . * Export mediation paths table
171 . esttab a_PS_T2 a_PSI_T2 a_EA_T2 a_SP_T2 outcome_model ///
    >     using "results/table4_H2_mediation_paths.csv", replace ///
    >     star(* 0.10 ** 0.05 *** 0.01) ///
    >     b(%9.3f) se(%9.3f) ///
    >     title("Table 4: H2 — Mediation a-paths and Outcome Model") ///
    >     mtitles("a: PS" "a: PSI" "a: EA" "a: SP" "Outcome") ///
    >     note("Robust standard errors in parentheses")
    (file results/table4_H2_mediation_paths.csv not found)
    (output written to results/table4_H2_mediation_paths.csv)

172 .
173 . * 5.3 Bootstrap indirect effects
174 . capture program drop boot_mediation

175 . program define boot_mediation, rclass
      1.     * a-paths
176 .     quietly regress PS_T2 Exposure_T2 IPB1 IBT_mean SC_mean age i.gender
      2.     local a1 = _b[Exposure_T2]
      3.     quietly regress PSI_T2 Exposure_T2 IPB1 IBT_mean SC_mean age i.gender
      4.     local a2 = _b[Exposure_T2]
      5.     quietly regress EA_T2 Exposure_T2 IPB1 IBT_mean SC_mean age i.gender
      6.     local a3 = _b[Exposure_T2]
      7.     quietly regress SP_T2 Exposure_T2 IPB1 IBT_mean SC_mean age i.gender
      8.     local a4 = _b[Exposure_T2]
      9.
177 .     * Outcome model (b-paths)
178 .     quietly regress IPB1_T2 Exposure_T2 PS_T2 PSI_T2 EA_T2 SP_T2 ///
    >                     IPB1 IBT_mean SC_mean age i.gender MA_T2_mean TA_T2_mean
    >  MOOD1_T2
     10.     local b1 = _b[PS_T2]
     11.     local b2 = _b[PSI_T2]
     12.     local b3 = _b[EA_T2]
     13.     local b4 = _b[SP_T2]
     14.     local cprime = _b[Exposure_T2]
     15.
```

```
179 .     * Indirect effects
180 .     return scalar ind_ps = `a1' * `b1'
     16.     return scalar ind_psi = `a2' * `b2'
     17.     return scalar ind_ea = `a3' * `b3'
     18.     return scalar ind_sp = `a4' * `b4'
     19.     return scalar total_indirect = `a1'*`b1' + `a2'*`b2' + `a3'*`b3' + `a
    > 4'*`b4'
     20.     return scalar direct = `cprime'
     21.     return scalar total = `cprime' + `a1'*`b1' + `a2'*`b2' + `a3'*`b3' +
    > `a4'*`b4'
     22.
181 .     * Contrasts
182 .     return scalar contrast_ps_psi = `a1'*`b1' - `a2'*`b2'
     23.     return scalar contrast_ps_ea = `a1'*`b1' - `a3'*`b3'
     24.     return scalar contrast_psi_sp = `a2'*`b2' - `a4'*`b4'
     25. end

183 .
184 . bootstrap r(ind_ps) r(ind_psi) r(ind_ea) r(ind_sp) ///
    >           r(total_indirect) r(direct) r(total) ///
    >           r(contrast_ps_psi) r(contrast_ps_ea) r(contrast_psi_sp), ///
    >           reps(5000) seed(2026): boot_mediation
    (running boot_mediation on estimation sample)

    Bootstrap replications (5,000): .........10.........20.........30.........40..
    > .......50.........60.........70.........80.........90.........100.........11
    > 0.........120.........130.........140.........150.........160.........170...
    > ......180.........190.........200.........210.........220.........230.......
    > ..240.........250.........260.........270.........280.........290.........30
    > 0.........310.........320.........330.........340.........350.........360...
    > ......370.........380.........390.........400.........410.........420.......
    > ..430.........440.........450.........460.........470.........480.........49
    > 0.........500.........510.........520.........530.........540.........550...
    > ......560.........570.........580.........590.........600.........610.......
    > ..620.........630.........640.........650.........660.........670.........68
    > 0.........690.........700.........710.........720.........730.........740...
    > ......750.........760.........770.........780.........790.........800.......
    > ..810.........820.........830.........840.........850.........860.........87
    > 0.........880.........890.........900.........910.........920.........930...
    > ......940.........950.........960.........970.........980.........990.......
    > ..1,000.........1,010.........1,020.........1,030.........1,040.........1,05
    > 0.........1,060.........1,070.........1,080.........1,090.........1,100.....
    > ....1,110.........1,120.........1,130.........1,140.........1,150.........1,
    > 160.........1,170.........1,180.........1,190.........1,200.........1,210...
    > ......1,220.........1,230.........1,240.........1,250.........1,260.........
    > 1,270.........1,280.........1,290.........1,300.........1,310.........1,320.
```

> ....1,330.........1,340.........1,350.........1,360.........1,370.......
> ..1,380.........1,390.........1,400.........1,410.........1,420.........1,43
> 0.........1,440.........1,450.........1,460.........1,470.........1,480.....
> ....1,490.........1,500.........1,510.........1,520.........1,530.........1,
> 540.........1,550.........1,560.........1,570.........1,580.........1,590...
> ......1,600.........1,610.........1,620.........1,630.........1,640.........
> 1,650.........1,660.........1,670.........1,680.........1,690.........1,700.
> ....1,710.........1,720.........1,730.........1,740.........1,750.......
> ..1,760.........1,770.........1,780.........1,790.........1,800.........1,81
> 0.........1,820.........1,830.........1,840.........1,850.........1,860.....
> ....1,870.........1,880.........1,890.........1,900.........1,910.........1,
> 920.........1,930.........1,940.........1,950.........1,960.........1,970...
> ......1,980.........1,990.........2,000.........2,010.........2,020.........
> 2,030.........2,040.........2,050.........2,060.........2,070.........2,080.
> ....2,090.........2,100.........2,110.........2,120.........2,130.......
> ..2,140.........2,150.........2,160.........2,170.........2,180.........2,19
> 0.........2,200.........2,210.........2,220.........2,230.........2,240.....
> ....2,250.........2,260.........2,270.........2,280.........2,290.........2,
> 300.........2,310.........2,320.........2,330.........2,340.........2,350...
> ......2,360.........2,370.........2,380.........2,390.........2,400.........
> 2,410.........2,420.........2,430.........2,440.........2,450.........2,460.
> ....2,470.........2,480.........2,490.........2,500.........2,510.......
> ..2,520.........2,530.........2,540.........2,550.........2,560.........2,57
> 0.........2,580.........2,590.........2,600.........2,610.........2,620.....
> ....2,630.........2,640.........2,650.........2,660.........2,670.........2,
> 680.........2,690.........2,700.........2,710.........2,720.........2,730...
> ......2,740.........2,750.........2,760.........2,770.........2,780.........
> 2,790.........2,800.........2,810.........2,820.........2,830.........2,840.
> ....2,850.........2,860.........2,870.........2,880.........2,890.......
> ..2,900.........2,910.........2,920.........2,930.........2,940.........2,95
> 0.........2,960.........2,970.........2,980.........2,990.........3,000.....
> ....3,010.........3,020.........3,030.........3,040.........3,050.........3,
> 060.........3,070.........3,080.........3,090.........3,100.........3,110...
> ......3,120.........3,130.........3,140.........3,150.........3,160.........
> 3,170.........3,180.........3,190.........3,200.........3,210.........3,220.
> ....3,230.........3,240.........3,250.........3,260.........3,270.......
> ..3,280.........3,290.........3,300.........3,310.........3,320.........3,33
> 0.........3,340.........3,350.........3,360.........3,370.........3,380.....
> ....3,390.........3,400.........3,410.........3,420.........3,430.........3,
> 440.........3,450.........3,460.........3,470.........3,480.........3,490...
> ......3,500.........3,510.........3,520.........3,530.........3,540.........
> 3,550.........3,560.........3,570.........3,580.........3,590.........3,600.
> ....3,610.........3,620.........3,630.........3,640.........3,650.......
> ..3,660.........3,670.........3,680.........3,690.........3,700.........3,71
> 0.........3,720.........3,730.........3,740.........3,750.........3,760.....
> ....3,770.........3,780.........3,790.........3,800.........3,810.........3,

```
> 820.........3,830.........3,840.........3,850.........3,860.........3,870...
> ......3,880.........3,890.........3,900.........3,910.........3,920.........
> 3,930.........3,940.........3,950.........3,960.........3,970.........3,980.
> ........3,990.........4,000.........4,010.........4,020.........4,030.......
> ..4,040.........4,050.........4,060.........4,070.........4,080.........4,09
> 0.........4,100.........4,110.........4,120.........4,130.........4,140.....
> ....4,150.........4,160.........4,170.........4,180.........4,190.........4,
> 200.........4,210.........4,220.........4,230.........4,240.........4,250...
> ......4,260.........4,270.........4,280.........4,290.........4,300.........
> 4,310.........4,320.........4,330.........4,340.........4,350.........4,360.
> ........4,370.........4,380.........4,390.........4,400.........4,410.......
> ..4,420.........4,430.........4,440.........4,450.........4,460.........4,47
> 0.........4,480.........4,490.........4,500.........4,510.........4,520.....
> ....4,530.........4,540.........4,550.........4,560.........4,570.........4,
> 580.........4,590.........4,600.........4,610.........4,620.........4,630...
> ......4,640.........4,650.........4,660.........4,670.........4,680.........
> 4,690.........4,700.........4,710.........4,720.........4,730.........4,740.
> ........4,750.........4,760.........4,770.........4,780.........4,790.......
> ..4,800.........4,810.........4,820.........4,830.........4,840.........4,85
> 0.........4,860.........4,870.........4,880.........4,890.........4,900.....
> ....4,910.........4,920.........4,930.........4,940.........4,950.........4,
> 960.........4,970.........4,980.........4,990.........5,000 done

Bootstrap results                                        Number of obs =   219
                                                         Replications  = 5,000

      Command: boot_mediation
        _bs_1: r(ind_ps)
        _bs_2: r(ind_psi)
        _bs_3: r(ind_ea)
        _bs_4: r(ind_sp)
        _bs_5: r(total_indirect)
        _bs_6: r(direct)
        _bs_7: r(total)
        _bs_8: r(contrast_ps_psi)
        _bs_9: r(contrast_ps_ea)
       _bs_10: r(contrast_psi_sp)
```

	Observed coefficient	Bootstrap std. err.	z	P>|z|	Normal-based [95% conf.	interval]
_bs_1	.1552252	.1023085	1.52	0.129	-.0452957	.3557461
_bs_2	.1384862	.1264435	1.10	0.273	-.1093385	.3863108
_bs_3	-.0027776	.1448548	-0.02	0.985	-.2866877	.2811325
_bs_4	.2358883	.1040065	2.27	0.023	.0320393	.4397373
_bs_5	.5268221	.2789317	1.89	0.059	-.019874	1.073518
_bs_6	.6101052	.5365336	1.14	0.255	-.4414813	1.661692
_bs_7	1.136927	.4817583	2.36	0.018	.1926984	2.081156
_bs_8	.016739	.1510982	0.11	0.912	-.2794079	.312886
_bs_9	.1580028	.1613173	0.98	0.327	-.1581733	.474179
_bs_10	-.0974021	.1512883	-0.64	0.520	-.3939218	.1991176

```
185 .
186 . estat bootstrap, percentile

    Bootstrap results                               Number of obs     =        219
                                                    Replications      =       5000

         Command: boot_mediation
           _bs_1: r(ind_ps)
           _bs_2: r(ind_psi)
           _bs_3: r(ind_ea)
           _bs_4: r(ind_sp)
           _bs_5: r(total_indirect)
           _bs_6: r(direct)
           _bs_7: r(total)
           _bs_8: r(contrast_ps_psi)
           _bs_9: r(contrast_ps_ea)
          _bs_10: r(contrast_psi_sp)
```

	Observed coefficient	Bias	Bootstrap std. err.	[95% conf.	interval]	
_bs_1	.15522521	.0014524	.10230848	-.0143676	.383408	(P)
_bs_2	.13848619	.0013135	.12644347	-.097634	.4075078	(P)
_bs_3	-.00277764	-.0024551	.14485475	-.292589	.280487	(P)
_bs_4	.23588829	.0000587	.10400651	.0574231	.4668502	(P)
_bs_5	.52682205	.0003694	.27893168	.0223265	1.108142	(P)
_bs_6	.61010518	-.0301959	.53653357	-.4719979	1.619341	(P)
_bs_7	1.1369272	-.0298265	.48175825	.1935693	2.062646	(P)
_bs_8	.01673902	.0001389	.15109816	-.2825456	.3199203	(P)

```
       _bs_9 |   .15800284   .0039075   .16131731   -.1497393   .4886567   (P)
      _bs_10 |   -.0974021   .0012548   .15128834   -.4040715   .1999787   (P)
------------------------------------------------------------------------------
    Key: P: Percentile

187 .
188 . * Export bootstrap results
189 . esttab . using "results/table5_H2_bootstrap_indirect.csv", replace ///
>     ci(%9.4f) wide ///
>     title("Table 5: H2 — Bootstrap Indirect Effects (5000 reps)")
(file results/table5_H2_bootstrap_indirect.csv not found)
(output written to results/table5_H2_bootstrap_indirect.csv)

190 .
191 . /*------------------------------------------------------------------------
>   SECTION 6: H3a-b — MODERATED MEDIATION
> ----------------------------------------------------------------------*/
192 .
193 . display _n "=== H3a: IBT moderates b-paths ==="

=== H3a: IBT moderates b-paths ===

194 .
195 . * 6.1 H3a: IBT moderates mediator → IPB paths
196 . regress IPB1_T2 c.Exposure_T2 ///
>         c.PS_T2##c.IBT_c c.PSI_T2##c.IBT_c ///
>         c.EA_T2##c.IBT_c c.SP_T2##c.IBT_c ///
>         IPB1 SC_mean age i.gender ///
>         MA_T2_mean TA_T2_mean MOOD1_T2, ///
>         vce(robust)
note: IBT_c omitted because of collinearity.
note: IBT_c omitted because of collinearity.
note: IBT_c omitted because of collinearity.

Linear regression                               Number of obs     =        219
                                                F(18, 200)        =       2.21
                                                Prob > F          =     0.0042
                                                R-squared         =     0.1515
                                                Root MSE          =     4.7543
```

```
────────────────────┬──────────────────────────────────────────────────────────────
> ─────
                    │               Robust
            IPB1_T2 │ Coefficient  std. err.      t    P>|t|     [95% conf. inter
> val]
────────────────────┼──────────────────────────────────────────────────────────────
> ─────
        Exposure_T2 │   .6202324   .5223221     1.19   0.236    -.4097325    1.65
> 0197
              PS_T2 │   .4683976   .2737592     1.71   0.089    -.0714272    1.00
> 8222
              IBT_c │  -.4448726   2.443949    -0.18   0.856    -5.264087    4.37
> 4341
                    │
  c.PS_T2#c.IBT_c │   .0185613   .2739551     0.07   0.946    -.5216496    .558
> 7722
                    │
             PSI_T2 │   .3602658   .3155813     1.14   0.255    -.2620278    .982
> 5594
              IBT_c │          0  (omitted)
                    │
 c.PSI_T2#c.IBT_c │   .0311207   .4233693     0.07   0.941    -.8037196     .86
> 5961
                    │
              EA_T2 │      .0175   .3502171     0.05   0.960    -.6730917    .708
> 0917
              IBT_c │          0  (omitted)
                    │
  c.EA_T2#c.IBT_c │  -.1547348   .3741805    -0.41   0.680      -.89258    .583
> 1104
                    │
              SP_T2 │   .6988448   .2754896     2.54   0.012     .1556079    1.24
> 2082
              IBT_c │          0  (omitted)
                    │
  c.SP_T2#c.IBT_c │   .0199481   .3799019     0.05   0.958     -.729179    .769
> 0753
                    │
               IPB1 │   .2559115   .0952332     2.69   0.008     .0681216    .443
> 7014
            SC_mean │  -.4775651   .4570699    -1.04   0.297     -1.37886    .423
> 7294
                age │   .0513434   .0929462     0.55   0.581    -.1319368    .234
> 6236
                    │
             gender │
```

```
               1  |  -.5659101   .9547227    -0.59   0.554    -2.448524    1.31
> 6704
               2  |  -.8091433    .865425    -0.93   0.351    -2.515672     .89
> 7385
                  |
      MA_T2_mean  |   .5491392   .3119933     1.76   0.080    -.0660791    1.16
> 4358
      TA_T2_mean  |  -.1034983   .2799437    -0.37   0.712    -.6555182    .448
> 5216
        MOOD1_T2  |  -.1766752   .3232377    -0.55   0.585    -.8140664    .460
> 7161
           _cons  |  -2.543047   4.201782    -0.61   0.546    -10.82852     5.7
> 4243
-----------------------------------------------------------------------------
> ----

197 . estimates store h3a

198 .
199 . * Probe interactions at +/- 1 SD IBT
200 . summarize IBT_c

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
       IBT_c |        219   -5.24e-09    .8544658  -2.021626   2.422818

201 . local ibt_lo = -r(sd)

202 . local ibt_hi = r(sd)

203 . margins, dydx(PS_T2) at(IBT_c = (`ibt_lo' `ibt_hi'))

Average marginal effects                                Number of obs = 219
Model VCE: Robust

Expression: Linear prediction, predict()
dy/dx wrt:  PS_T2
1._at: IBT_c = -.8544658
2._at: IBT_c =  .8544658
```

	dy/dx	Delta-method std. err.	t	P>|t|	[95% conf. interval]	
PS_T2						
_at						
1	**.4525376**	**.4002859**	**1.13**	**0.260**	**-.3367846**	**1.24186**
2	**.4842576**	**.3150416**	**1.54**	**0.126**	**-.1369718**	**1.105487**

```
204 . margins, dydx(PSI_T2) at(IBT_c = (`ibt_lo' `ibt_hi'))
```

Average marginal effects Number of obs = **219**
Model VCE: **Robust**

Expression: **Linear prediction, predict()**
dy/dx wrt: **PSI_T2**
1._at: IBT_c = **-.8544658**
2._at: IBT_c = **.8544658**

	dy/dx	Delta-method std. err.	t	P>|t|	[95% conf. interval]	
PSI_T2						
_at						
1	**.3336742**	**.5371702**	**0.62**	**0.535**	**-.7255697**	**1.392918**
2	**.3868574**	**.4151674**	**0.93**	**0.353**	**-.4318097**	**1.205525**

```
205 .
206 . display _n "=== H3b: Self-Control moderates b-paths ==="
```

=== H3b: Self-Control moderates b-paths ===

```
207 .
208 . * 6.2 H3b: SC moderates mediator → IPB paths
209 . regress IPB1_T2 c.Exposure_T2 ///
    >         c.PS_T2##c.SC_c c.PSI_T2##c.SC_c ///
    >         c.EA_T2##c.SC_c c.SP_T2##c.SC_c ///
    >         IPB1 IBT_mean age i.gender ///
    >         MA_T2_mean TA_T2_mean MOOD1_T2, ///
    >         vce(robust)
    note: SC_c omitted because of collinearity.
    note: SC_c omitted because of collinearity.
    note: SC_c omitted because of collinearity.

    Linear regression                               Number of obs     =        219
                                                    F(18, 200)        =       2.27
                                                    Prob > F          =     0.0031
                                                    R-squared         =     0.1612
                                                    Root MSE          =     4.4496

    ────────────────┬──────────────────────────────────────────────────────────────
    > ───
                    │               Robust
            IPB1_T2 │ Coefficient  std. err.      t    P>|t|     [95% conf. interv
    > al]
    ────────────────┼──────────────────────────────────────────────────────────────
    > ───
        Exposure_T2 │   .6152894   .5474632     1.12   0.262    -.4642513     1.69
    > 483
              PS_T2 │   .5275531   .2878558     1.83   0.068    -.0400686    1.095
    > 175
               SC_c │   3.129009    2.28194     1.37   0.172    -1.370741    7.628
    > 759
                    │
     c.PS_T2#c.SC_c │  -.4042036    .320986    -1.26   0.209    -1.037155    .2287
    > 474
                    │
             PSI_T2 │   .3625376   .3290964     1.10   0.272    -.2864064    1.011
    > 482
               SC_c │          0  (omitted)
                    │
    c.PSI_T2#c.SC_c │  -.1355901   .3094765    -0.44   0.662    -.7458457    .4746
    > 654
                    │
              EA_T2 │  -.0438786   .3257254    -0.13   0.893    -.6861752     .598
    > 418
               SC_c │          0  (omitted)
                    │
```

```
c.EA_T2#c.SC_c |  -.2288752    .288207    -0.79   0.428    -.7971895     .339
> 439
               |
         SP_T2 |   .7132407   .2742255     2.60   0.010     .1724965    1.253
> 985
          SC_c |          0  (omitted)
               |
c.SP_T2#c.SC_c |  -.0906289   .3423598    -0.26   0.791     -.765727    .5844
> 692
               |
          IPB1 |   .2529625   .0960082     2.63   0.009     .0636443    .4422
> 807
      IBT_mean |  -.7636267   .5607154    -1.36   0.175    -1.869299    .3420
> 459
           age |   .0608304    .097556     0.62   0.534    -.1315399    .2532
> 008
               |
        gender |
            1  |    -.61804   .9685393    -0.64   0.524    -2.527899    1.291
> 819
            2  |  -.8367179   .8469397    -0.99   0.324    -2.506795    .8333
> 593
               |
    MA_T2_mean |    .512567   .2878422     1.78   0.076     -.055028    1.080
> 162
    TA_T2_mean |  -.1067386   .2954634    -0.36   0.718    -.6893617    .4758
> 846
      MOOD1_T2 |  -.1702466   .3147994    -0.54   0.589    -.7909984    .4505
> 052
         _cons |  -1.557569   4.429169    -0.35   0.725    -10.29143    7.176
> 293
-------------------------------------------------------------------------------
> ---
```

```
210 . estimates store h3b

211 .
212 . * Export moderation table
213 . esttab h3a h3b ///
    >     using "results/table6_H3_moderation.csv", replace ///
    >     star(* 0.10 ** 0.05 *** 0.01) ///
    >     b(%9.3f) se(%9.3f) ///
    >     title("Table 6: H3 — Moderated Mediation") ///
    >     mtitles("H3a: IBT moderator" "H3b: SC moderator") ///
    >     note("Robust standard errors in parentheses")
    (file results/table6_H3_moderation.csv not found)
    (output written to results/table6_H3_moderation.csv)

214 .
215 . /*-----------------------------------------------------------------------
    >   SECTION 7: ROBUSTNESS — PROPENSITY SCORE MATCHING
    > -----------------------------------------------------------------------*/
216 .
217 . display _n "=== Robustness: PSM ==="

    === Robustness: PSM ===

218 .
219 . * 7.1 PSM — Nearest-Neighbor Matching
220 . * NOTE: With n=219, all demographics treated as ordinal (not i.) to
221 . *       maintain propensity-score overlap and avoid sparse-cell issues.
222 . *       Caliper removed — nn(1) guarantees a match; balance checked below.
223 . teffects psmatch (IPB1_T2) ///
    >     (Exposure_binary IBT_mean SC_mean SE_mean SII_mean ///
    >      age gender education occupation income relationship IPB1 FOMO_mean), //
    > /
    >     atet nn(1)

    Treatment-effects estimation                   Number of obs      =        217
    Estimator      : propensity-score matching     Matches: requested =          1
    Outcome model  : matching                                    min =          1
    Treatment model: logit                                       max =          1
    ─────────────────┬──────────────────────────────────────────────────────────────
    > ───
                     │               AI robust
             IPB1_T2 │ Coefficient  std. err.      z    P>|z|     [95% conf. interv
    > al]
    ─────────────────┼──────────────────────────────────────────────────────────────
    > ───
    ATET             │
```

```
Exposure_binary |
      (1 vs 0)  |   1.314815   .7975529     1.65   0.099    -.2483602     2.87
> 799
                |
----------------+--------------------------------------------------------------
> ---

224 . estimates store psm_att

225 .
226 . * Balance diagnostics
227 . tebalance summarize
    (refitting the model using the generate() option)

    Covariate balance summary

                             Raw      Matched
    -----------------------------------------
    Number of obs =          217          216
    Treated obs   =          108          108
    Control obs   =          109          108
    -----------------------------------------

    -----------------------------------------------------------------
                    |Standardized differences          Variance ratio
                    |        Raw     Matched           Raw    Matched
    ----------------+------------------------------------------------
           IBT_mean |   .5631549    -.047961      1.008353   .9129182
            SC_mean |  -.0674381   -.0255285       1.12555   1.432348
            SE_mean |   .5458658    .2054467      .9092237    1.08452
           SII_mean |   .3288875     .030456       1.14803   1.073678
                age |   .3529233   -.0591586      1.261411   .9990114
             gender |   .2044287   -.1515508      .9422974   1.436741
          education |    .042321   -.0589828      .8520581   1.054097
         occupation |   .2141833   -.0076631      1.773849   2.740909
             income |  -.0501173    .1276898      .7653198   2.908397
       relationship |  -.1678035    .0925277      .7898167   .9145221
               IPB1 |   .1586984    .1751355      1.619234   1.427675
          FOMO_mean |    .114304   -.0312665      .9173052   .7006495
    -----------------------------------------------------------------
```

```
228 . tebalance density
    (refitting the model using the generate() option)

229 . graph export "results/fig2_psm_balance_density.png", replace width(1200)
    file
        /Users/mhdquan/Documents/thesis-data-analytics/results/fig2_psm_balance_
        > density.png saved as PNG format

230 .
231 . * 7.2 Alternative: Inverse Probability Weighting (IPW)
232 . * More robust than PSM with limited overlap; does not require 1:1 matches
233 . teffects ipw (IPB1_T2) ///
    >     (Exposure_binary IBT_mean SC_mean SE_mean SII_mean ///
    >      age gender education occupation income relationship IPB1 FOMO_mean), //
    > /
    >     atet

    Iteration 0:  EE criterion = 4.401e-23
    Iteration 1:  EE criterion = 9.003e-31

    Treatment-effects estimation                    Number of obs     =        217
    Estimator      : inverse-probability weights
    Outcome model  : weighted mean
    Treatment model: logit
    ────────────────┬──────────────────────────────────────────────────────────────
    > ───
                    │               Robust
            IPB1_T2 │ Coefficient  std. err.      z    P>|z|     [95% conf. interv
    > al]
    ────────────────┼──────────────────────────────────────────────────────────────
    > ───
    ATET            │
    Exposure_binary │
          (1 vs 0)  │   1.309874   .7153213     1.83   0.067    -.0921295    2.711
    > 878
    ────────────────┼──────────────────────────────────────────────────────────────
    > ───
    POmean          │
    Exposure_binary │
                0   │   3.958644   .5394713     7.34   0.000       2.9013    5.015
    > 988
    ────────────────┴──────────────────────────────────────────────────────────────
    > ───
```

```
234 . estimates store ipw_att

235 .
236 . * Compare PSM vs. IPW estimates
237 . estimates table psm_att ipw_att, star(0.10 0.05 0.01)
```

Variable	psm_att	ipw_att
ATET		
Exposure_b~y		
(1 vs 0)	**1.3148148***	**1.3098745***
POmean		
0.Exposure~y		**3.9586441*****
TME1		
IBT_mean		**.5720605****
SC_mean		**.30545023**
SE_mean		**.5373537*****
SII_mean		**.27216755**
age		**.10606905***
gender		**.23670578**
education		**-.01665459**
occupation		**.18811928**
income		**-.00999866**
relationship		**-.23331685**
IPB1		**.00393709**
FOMO_mean		**-.17820536**
_cons		**-9.5647069*****

Legend: * p<.1; ** p<.05; *** p<.01

```
238 .
```

```
239 . esttab psm_att ipw_att ///
    >     using "results/table7_robustness_PSM_IPW.csv", replace ///
    >     star(* 0.10 ** 0.05 *** 0.01) ///
    >     b(%9.3f) se(%9.3f) ///
    >     title("Table 7: Robustness — PSM and IPW Estimates") ///
    >     mtitles("PSM (nn=1)" "IPW") ///
    >     note("ATET estimates")
    (file results/table7_robustness_PSM_IPW.csv not found)
    (output written to results/table7_robustness_PSM_IPW.csv)

240 .
241 . * 7.3 Generalized Propensity Score (GPS) — Continuous Treatment
242 . * Hirano & Imbens (2004): uses continuous Exposure_T2 directly,
243 . * avoiding information loss from dichotomization.
244 . display _n "=== GPS: Continuous Exposure ==="

    === GPS: Continuous Exposure ===

245 .
246 . * Step 1: Treatment model — E[Exposure_T2 | X]
247 . quietly regress Exposure_T2 IBT_mean SC_mean SE_mean SII_mean ///
    >     age gender education occupation income relationship IPB1 FOMO_mean

248 . predict double gps_mu, xb
    (2 missing values generated)

249 . local gps_sigma = e(rmse)

250 . display "Treatment model R² = " %5.3f e(r2) "  σ = " %5.3f `gps_sigma'
    Treatment model R² = 0.232  σ = 0.810

251 .
252 . * Step 2: Compute GPS = φ(T; μ(X), σ)
253 . gen double gps = normalden(Exposure_T2, gps_mu, `gps_sigma')
    (2 missing values generated)
```

```
254 . label var gps "Generalized Propensity Score"

255 . summarize gps, detail

                   Generalized Propensity Score
-------------------------------------------------------------
      Percentiles      Smallest
 1%     .0416207       .0026575
 5%     .1132941       .0344425
10%     .1304465       .0416207       Obs                 217
25%     .2512249       .0489544       Sum of wgt.         217

50%     .3973255                      Mean           .3500522
                        Largest       Std. dev.      .1345121
75%      .466325       .4924055
90%     .4877656       .4924848       Variance       .0180935
95%     .4912627       .4925238       Skewness      -.7296128
99%     .4924848       .4925266       Kurtosis       2.234595

256 .
257 . * Step 3: GPS-adjusted outcome model (dose-response)
258 . * Without GPS (naïve)
259 . regress IPB1_T2 Exposure_T2 ///
>     IPB1 IBT_mean SC_mean SE_mean age gender ///
>     MA_T2_mean TA_T2_mean MOOD1_T2, vce(robust)

Linear regression                               Number of obs     =        219
                                                F(10, 208)        =       3.12
                                                Prob > F          =     0.0010
                                                R-squared         =     0.1094
                                                Root MSE          =      4.496
```

IPB1_T2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
Exposure_T2	1.202231	.4721248	2.55	0.012	.2714679	2.132994
IPB1	.2584322	.0925731	2.79	0.006	.0759304	.440934
IBT_mean	-.6786984	.606227	-1.12	0.264	-1.873835	.5164385
SC_mean	-.42775	.4531989	-0.94	0.346	-1.321202	.465702
SE_mean	-.2446554	.309406	-0.79	0.430	-.8546291	.3653183
age	.041491	.0885876	0.47	0.640	-.1331537	.2161356
gender	-.3455469	.3961788	-0.87	0.384	-1.126588	.4354937
MA_T2_mean	.5606625	.2807494	2.00	0.047	.0071834	1.114141
TA_T2_mean	-.0710338	.2762153	-0.26	0.797	-.6155743	.4735067
MOOD1_T2	-.049813	.3197476	-0.16	0.876	-.6801745	.5805485

```
         _cons |   6.786268   3.654013     1.86   0.065    -.4173793    13.98991
------------------------------------------------------------------------------

260 . estimates store gps_naive

261 .
262 . * With GPS control (bias-adjusted)
263 . regress IPB1_T2 Exposure_T2 gps c.Exposure_T2#c.gps ///
    >     IPB1 IBT_mean SC_mean SE_mean age gender ///
    >     MA_T2_mean TA_T2_mean MOOD1_T2, vce(robust)

    Linear regression                               Number of obs     =        217
                                                    F(12, 204)        =       2.99
                                                    Prob > F          =     0.0007
                                                    R-squared         =     0.1343
                                                    Root MSE          =     4.4557

    --------------------------------------------------------------------------
    > -------
                       |               Robust
               IPB1_T2 | Coefficient  std. err.      t    P>|t|     [95% conf. in
    > terval]
    -------------------+------------------------------------------------------
    > -------
           Exposure_T2 |   1.242836   .9970186     1.25   0.214    -.7229468    3
    > .208618
                   gps |  -5.586162   2.650819    -2.11   0.036    -10.81268   -.
    > 3596455
                       |
    c.Exposure_T2#c.gps |  -.9940158      3.326    -0.30   0.765    -7.551761    5
    > .563729
                       |
                  IPB1 |   .2805991   .0936633     3.00   0.003     .0959269    .
    > 4652714
              IBT_mean |  -.6304753   .5520476    -1.14   0.255    -1.718926    .
    > 4579754
               SC_mean |  -.4181444   .4423735    -0.95   0.346    -1.290355    .
    > 4540661
               SE_mean |  -.1475397   .3603528    -0.41   0.683    -.8580333    .
    > 5629538
                   age |   .0672375    .093901     0.72   0.475    -.1179033    .
    > 2523784
                gender |   -.412417   .4064202    -1.01   0.311     -1.21374    .
    > 3889057
            MA_T2_mean |   .5545405   .2826171     1.96   0.051    -.0026846    1
    > .111766
```

```
        TA_T2_mean |   -.199821    .268716    -0.74   0.458    -.7296378    .
> 3299958
          MOOD1_T2 |  -.0274652   .2880679    -0.10   0.924    -.5954375    .
> 5405071
             _cons |   7.784034   3.792451     2.05   0.041     .3066071    1
> 5.26146
-------------------+----------------------------------------------------------
> --------

264 . estimates store gps_adjusted

265 .
266 . * Compare: effect of Exposure_T2 should be consistent
267 . estimates table gps_naive gps_adjusted, ///
>     star(0.10 0.05 0.01) stats(N r2) keep(Exposure_T2)

----------------------------------------------
    Variable |  gps_naive     gps_adjusted
-------------+--------------------------------
 Exposure_T2 |  1.202231**     1.2428358
-------------+--------------------------------
           N |       219           217
          r2 |  .1093676      .13426113
----------------------------------------------
           Legend: * p<.1; ** p<.05; *** p<.01

268 .
269 . esttab gps_naive gps_adjusted ///
>     using "results/table8_robustness_GPS.csv", replace ///
>     star(* 0.10 ** 0.05 *** 0.01) ///
>     b(%9.3f) se(%9.3f) scalars("r2 R-squared") ///
>     title("Table 8: Robustness - GPS Dose-Response") ///
>     mtitles("Without GPS" "GPS-adjusted") ///
>     note("Robust SE; GPS = Generalized Propensity Score (Hirano & Imbens 200
> 4)")
(file results/table8_robustness_GPS.csv not found)
(output written to results/table8_robustness_GPS.csv)
```

```
270 .
271 . * Marginal effect of exposure at mean GPS
272 . quietly estimates restore gps_adjusted

273 . margins, dydx(Exposure_T2) atmeans

    Conditional marginal effects                             Number of obs = 217
    Model VCE: Robust

    Expression: Linear prediction, predict()
    dy/dx wrt:  Exposure_T2
    At: Exposure_T2 = -.0040491 (mean)
        gps         =  .3500522 (mean)
        IPB1        =  3.262673 (mean)
        IBT_mean    =  3.793971 (mean)
        SC_mean     =  3.564546 (mean)
        SE_mean     =   4.56106 (mean)
        age         =  27.45622 (mean)
        gender      =   1.43318 (mean)
        MA_T2_mean  =  3.694316 (mean)
        TA_T2_mean  =  3.711214 (mean)
        MOOD1_T2    =  3.451613 (mean)

    ------------------------------------------------------------------------------
                 |            Delta-method
                 |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
    -------------+----------------------------------------------------------------
     Exposure_T2 |   .8948784   .4868997     1.84   0.068    -.0651227     1.85488
    ------------------------------------------------------------------------------

274 .
275 . * Clean up
276 . drop gps_mu gps
```

```
. 
. /*-------------------------------------------------------------------------
>   SECTION 8: ROBUSTNESS - E-VALUE SENSITIVITY ANALYSIS
> -------------------------------------------------------------------------*/
. 
. display _n "=== Robustness: E-value ==="

=== Robustness: E-value ===

. 
. * Compute E-value from primary regression coefficient
. * Using point estimate and CI lower bound from H1
. estimates restore h1_primary
(results h1_primary are active now)

. local b_exp = _b[Exposure_T2]

. local se_exp = _se[Exposure_T2]

. local ci_lo = `b_exp' - 1.96 * `se_exp'

. 
. * E-value formula (for continuous outcome, approximate via RR conversion)
. * RR ≈ exp(0.91 * beta / SD_Y) per VanderWeele & Ding (2017)
. quietly summarize IPB1_T2

. local sd_y = r(sd)

. local RR = exp(0.91 * `b_exp' / `sd_y')

. local E_point = `RR' + sqrt(`RR' * (`RR' - 1))

. local RR_lo = exp(0.91 * `ci_lo' / `sd_y')
```

```
. local E_ci = `RR_lo' + sqrt(`RR_lo' * (`RR_lo' - 1))

.
. display "E-value (point estimate): " %6.3f `E_point'
E-value (point estimate):  1.841

. display "E-value (CI lower bound): " %6.3f `E_ci'
E-value (CI lower bound):  1.291

.
. /*-----------------------------------------------------------------
>   SECTION 9: ATTRITION ANALYSIS
> -----------------------------------------------------------------*/
.
. display _n "=== Attrition Analysis ==="

=== Attrition Analysis ===

.
. * Load T1-only data with completed_t2 flag (generated by merge_panel.py)
. preserve

. use "t1_attrition.dta", clear

.
. display _n "Attrition summary:"

Attrition summary:

. tabulate completed_t2

completed_t |
          2 |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        278       55.94       55.94
          1 |        219       44.06      100.00
------------+-----------------------------------
      Total |        497      100.00
```

```
311 .
312 . * Compare completers vs. dropouts on key T1 variables
313 . foreach v in age IBT1 SC1 SE1 EXP5 IPB1 {
  2.     display _n "`v':"
  3.     ttest `v', by(completed_t2)
  4. }
```

age:

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
0	**278**	**20.76259**	**.1593077**	**2.65619**	**20.44898**	**21.0762**
1	**219**	**27.43836**	**.2023032**	**2.993815**	**27.03964**	**27.83708**
Combined	**497**	**23.70423**	**.1949366**	**4.345818**	**23.32122**	**24.08723**
diff		**-6.675766**	**.2538763**		**-7.174574**	**-6.176958**

```
    diff = mean(0) - mean(1)                                      t = -26.2953
H0: diff = 0                                     Degrees of freedom =      495

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000
```

IBT1:

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
0	**278**	**3.938849**	**.0771173**	**1.285802**	**3.787039**	**4.090659**
1	**219**	**3.995434**	**.0859873**	**1.272496**	**3.825961**	**4.164907**
Combined	**497**	**3.963783**	**.05737**	**1.278977**	**3.851065**	**4.076501**
diff		**-.0565849**	**.1156458**		**-.283802**	**.1706322**

```
    diff = mean(0) - mean(1)                                      t =  -0.4893
H0: diff = 0                                     Degrees of freedom =      495

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.3124         Pr(|T| > |t|) = 0.6248          Pr(T > t) = 0.6876
```

SC1:

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
0	278	4.053957	.0749161	1.249102	3.90648	4.201434
1	218	3.788991	.0867852	1.281369	3.617941	3.960041
Combined	496	3.9375	.0569774	1.268947	3.825553	4.049447
diff		.264966	.1142939		.0404038	.4895282

diff = mean(**0**) - mean(**1**) t = **2.3183**
H0: diff = 0 Degrees of freedom = **494**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = **0.9896** Pr(|T| > |t|) = **0.0208** Pr(T > t) = **0.0104**

SE1:

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
0	278	4.546763	.0712469	1.187924	4.406508	4.687017
1	219	4.484018	.0793619	1.174449	4.327603	4.640433
Combined	497	4.519115	.0529853	1.181228	4.415011	4.623218
diff		.0627443	.1067958		-.1470847	.2725733

diff = mean(**0**) - mean(**1**) t = **0.5875**
H0: diff = 0 Degrees of freedom = **495**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = **0.7214** Pr(|T| > |t|) = **0.5571** Pr(T > t) = **0.2786**

EXP5:

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
0	**278**	**3.003597**	**.0718011**	**1.197163**	**2.862252**	**3.144942**
1	**219**	**2.908676**	**.0858871**	**1.271013**	**2.7394**	**3.077951**
Combined	**497**	**2.961771**	**.0551685**	**1.229898**	**2.853378**	**3.070163**
diff		**.0949213**	**.111153**		**-.1234686**	**.3133112**

diff = mean(**0**) - mean(**1**) t = **0.8540**

H0: diff = 0 Degrees of freedom = **495**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0

Pr(T < t) = **0.8032** Pr(|T| > |t|) = **0.3935** Pr(T > t) = **0.1968**

IPB1:

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
0	**278**	**2.856115**	**.1829074**	**3.049676**	**2.49605**	**3.21618**
1	**219**	**3.246575**	**.2396832**	**3.546988**	**2.774182**	**3.718968**
Combined	**497**	**3.028169**	**.1471479**	**3.280442**	**2.739059**	**3.317279**
diff		**-.3904602**	**.2961715**		**-.9723685**	**.191448**

diff = mean(**0**) - mean(**1**) t = **-1.3184**

H0: diff = 0 Degrees of freedom = **495**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0

Pr(T < t) = **0.0940** Pr(|T| > |t|) = **0.1880** Pr(T > t) = **0.9060**

```
314 .
315 . tabulate gender completed_t2, chi2

           |     completed_t2
    gender |         0          1 |     Total
-----------+----------------------+----------
         0 |       211         33 |       244 
         1 |        23         57 |        80 
         2 |        44        129 |       173 
-----------+----------------------+----------
     Total |       278        219 |       497 

          Pearson chi2(2) = 181.6210   Pr = 0.000

316 . tabulate education completed_t2, chi2

           |     completed_t2
 education |         0          1 |     Total
-----------+----------------------+----------
         1 |        44         23 |        67 
         2 |        25         30 |        55 
         3 |       200        132 |       332 
         4 |         9         34 |        43 
-----------+----------------------+----------
     Total |       278        219 |       497 

          Pearson chi2(3) =  28.9025   Pr = 0.000

317 . restore

318 .
319 . /*----------------------------------------------------------------------
>   SECTION 10: COMMON METHOD VARIANCE
> ----------------------------------------------------------------------*/
```

```
320 .
321 . display _n "=== CMV: Marker Variable Check ==="
```

=== CMV: Marker Variable Check ===

```
322 .
323 . * Correlation of marker variable with focal constructs
324 . correlate MV1_T2 PS_T2 PSI_T2 EA_T2 SP_T2 IPB1_T2
(obs=219)
```

	MV1_T2	PS_T2	PSI_T2	EA_T2	SP_T2	IPB1_T2
MV1_T2	**1.0000**					
PS_T2	**-0.0041**	**1.0000**				
PSI_T2	**-0.0554**	**0.1343**	**1.0000**			
EA_T2	**-0.0548**	**0.1674**	**0.1556**	**1.0000**		
SP_T2	**0.0923**	**0.1841**	**0.0568**	**0.1827**	**1.0000**	
IPB1_T2	**-0.0748**	**0.1752**	**0.1529**	**0.0692**	**0.1962**	**1.0000**

```
325 .
326 . * Harman's single-factor test (T2 mediator items)
327 . factor PS1_T2 PS2_T2 PS3_T2 PS4_T2 PS5_T2 ///
>        PSI1_T2 PSI2_T2 PSI3_T2 PSI4_T2 PSI5_T2 ///
>        EA1_T2 EA2_T2 EA3_T2 EA4_T2 ///
>        SP1_T2 SP2_T2 SP3_T2 SP4_T2 ///
>        IPB4_T2 IPB5_T2 IPB6_T2, ///
>        factors(1) pcf
(obs=175)
```

```
Factor analysis/correlation                    Number of obs    =        175
    Method: principal-component factors        Retained factors =          1
    Rotation: (unrotated)                      Number of params =         21
```

Factor	Eigenvalue	Difference	Proportion	Cumulative
Factor1	**5.09877**	**2.11850**	**0.2428**	**0.2428**
Factor2	**2.98026**	**0.50485**	**0.1419**	**0.3847**
Factor3	**2.47541**	**0.39707**	**0.1179**	**0.5026**
Factor4	**2.07834**	**0.89464**	**0.0990**	**0.6016**
Factor5	**1.18370**	**0.44324**	**0.0564**	**0.6579**
Factor6	**0.74046**	**0.09160**	**0.0353**	**0.6932**
Factor7	**0.64886**	**0.04021**	**0.0309**	**0.7241**
Factor8	**0.60865**	**0.03878**	**0.0290**	**0.7531**
Factor9	**0.56987**	**0.03928**	**0.0271**	**0.7802**

Factor10	**0.53059**	**0.00961**	**0.0253**	**0.8055**
Factor11	**0.52097**	**0.04576**	**0.0248**	**0.8303**
Factor12	**0.47521**	**0.02291**	**0.0226**	**0.8529**
Factor13	**0.45230**	**0.02465**	**0.0215**	**0.8744**
Factor14	**0.42765**	**0.05017**	**0.0204**	**0.8948**
Factor15	**0.37748**	**0.00778**	**0.0180**	**0.9128**
Factor16	**0.36971**	**0.01474**	**0.0176**	**0.9304**
Factor17	**0.35497**	**0.05001**	**0.0169**	**0.9473**
Factor18	**0.30496**	**0.00730**	**0.0145**	**0.9618**
Factor19	**0.29766**	**0.02182**	**0.0142**	**0.9760**
Factor20	**0.27584**	**0.04750**	**0.0131**	**0.9891**
Factor21	**0.22834**	.	**0.0109**	**1.0000**

LR test: independent vs. saturated: chi2(**210**) = **1522.25** Prob>chi2 = **0.0000**

Factor loadings (pattern matrix) and unique variances

Variable	Factor1	Uniqueness
PS1_T2	**0.6273**	**0.6065**
PS2_T2	**0.5724**	**0.6723**
PS3_T2	**0.5355**	**0.7133**
PS4_T2	**0.5101**	**0.7398**
PS5_T2	**0.5643**	**0.6816**
PSI1_T2	**0.4546**	**0.7934**
PSI2_T2	**0.4579**	**0.7903**
PSI3_T2	**0.4303**	**0.8149**
PSI4_T2	**0.4856**	**0.7642**
PSI5_T2	**0.4313**	**0.8140**
EA1_T2	**0.4654**	**0.7834**
EA2_T2	**0.4157**	**0.8272**
EA3_T2	**0.4687**	**0.7804**
EA4_T2	**0.4416**	**0.8050**
SP1_T2	**0.3488**	**0.8784**
SP2_T2	**0.3745**	**0.8597**
SP3_T2	**0.4152**	**0.8276**
SP4_T2	**0.2187**	**0.9522**
IPB4_T2	**0.5908**	**0.6510**
IPB5_T2	**0.6255**	**0.6088**
IPB6_T2	**0.6801**	**0.5375**

```
328 . * Check: first factor should explain < 50% of total variance
329 .
330 . display _n "=== PIPELINE COMPLETE ==="

    === PIPELINE COMPLETE ===

331 . display "All commands executed successfully."
    All commands executed successfully.

332 . display "Results exported to: results/"
    Results exported to: results/

333 .
334 . * --- Close log ---
335 . log close main
          name:  main
           log:  /Users/mhdquan/Documents/thesis-data-analytics/results/analysis_l
    > og.smcl
      log type:  smcl
     closed on:  14 May 2026, 00:45:53
    ------------------------------------------------------------------------------
```